

Do completions follow permits later in Prague than in the rest of Czechia?

In official monthly counts, completions of flats in new Prague apartment buildings follow permits later than elsewhere in Czechia (supported); the best fit is 11 months longer, but anywhere from about 6 to over 30.

PRAGUE, HOMES PER 1 000 RESIDENTS	4.5
a year, 2015–2024 mean · administrative count, no sampling error	
PRAGUE METRO REGION, PER 1 000	5.28
a year, 2011–2021 census, which in Czechia includes reconstructions · 163rd of 208, median 3.55 · new houses only: 3.88, 116th, median 3.50	
PERMIT TO COMPLETION, PRAGUE – REST	+11.1 months
supported · best fit; size about 6 to 30+ months (profile likelihood); the bootstrap 8.7 to 12.6 understates it	
NEW FLATS, PER DOUBLING OF METRO DISTANCE	–0.23
log density, about 20 % fewer new flats per km ² · decision p 0.0252 against 0.025 (district bootstrap, about 14 clusters) · indeterminate	

OVERWHELMED? HERE'S THE SHORT VERSION

- Prague completes fewer homes per resident than Vienna or Warsaw; its region is in the upper quarter of European metropolitan regions on a census count that includes Czech reconstructions, and mid-table on new houses alone.
- After a permit, completions of flats in Prague's new apartment buildings are recorded later than elsewhere in Czechia (supported). The direction holds in every variant, but the size depends on the assumed lag shape, 41 months with an exponential lag, and the rest of Czechia's best-fitting lag lies at the edge of the grid searched.
- The gap starts at the permit, so it says nothing about how long getting a permit takes, and part of it may be when completions are registered or how large Prague's projects are.

- The metro test is indeterminate; the Polish builders' test and the demand and price estimates are uninformative.

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work since May 2024

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The question and what this adds

Housing economics distinguishes several reasons why a city adds few homes. Demand may be low; procedure may be slow, either in granting permits or in building once a permit is granted; or supply may not respond when prices rise, which happens where regulation binds and prices rather than quantities absorb demand^{2,3,4}. The responsiveness of supply to prices has been estimated from national series^{5,6}, and the Czech discussion of procedure has concentrated on how long a building permit takes, the step the 2021 building act addressed and the one the World Bank's discontinued Doing Business series measured for a stylised warehouse^{7,8}. This study measures the step after the permit with official monthly data, asking how long a permitted flat takes to be completed in Prague compared with the rest of Czechia, and places that measurement beside a census comparison of Prague with Europe's metropolitan regions, a test of where in Prague new flats are built, and a Polish test of whether non-market builders respond differently to prices.

Data

The starting point is a count that each country's statistical office publishes: new homes completed in a year for every thousand residents. Prague has 1.4 million residents, Vienna 2.0 million and Warsaw 1.9 million, and the three series come from the Czech Statistical Office, Statistik Austria and Statistics Poland^{9,10,11,12}.

Prague completes fewer homes per resident than Vienna or Warsaw, two high builders.

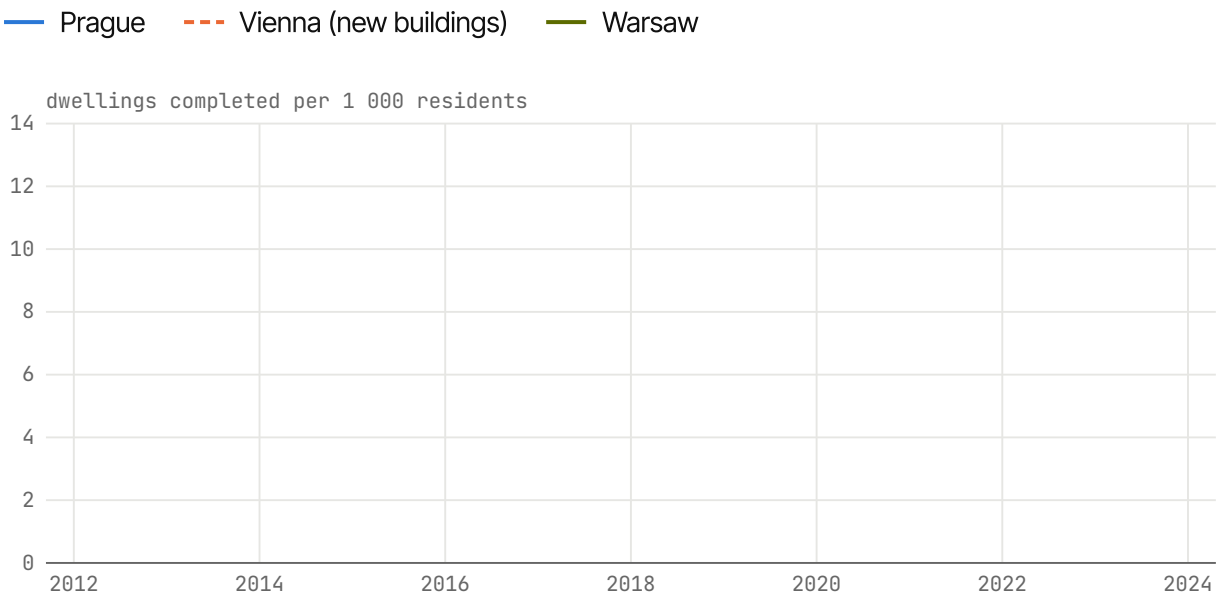


fig. 1 · Dwellings completed per 1 000 residents a year, 2012–2024, in Prague (blue), Vienna (orange, dashed) and Warsaw (olive); administrative counts, no sampling band. Prague and Warsaw count every completed home, including those made by extending or converting a building; Vienna's series counts homes in new buildings only, because conversions were recorded for Vienna only from 2024. With them, Vienna's 2024 figure is 6.2. Sources: ČSÚ, Statistik Austria, Statistics Poland, accessed 28 September 2026.

Prague's line is the flattest of the three and the lowest in every year but 2016. It runs between 3.1 homes per thousand residents in 2013 and 4.9 in 2022, and in 2024 the city completed 6 489 homes. Vienna climbed through the 2010s to 8.5 in 2021, when 16 262 homes were completed in new buildings, and fell to 5.4 in 2024, while Warsaw built more than both in every year, with a peak of 13.2 in 2018, or 23 430 homes. Over the ten years to 2024 Prague averaged 4.5, Vienna 6.6 and Warsaw 10.4. As the European comparison below shows, the metropolitan regions of both comparators build well above the European median, so these three lines set Prague beside two high builders.

Year	Prague	Vienna, new buildings	Warsaw
2012	3.2 (4 024)	3.6 (6 295)	7.9 (13 496)
2013	3.1 (3 844)	4.1 (7 124)	7.6 (13 128)
2014	3.9 (4 848)	5.0 (8 877)	8.7 (14 964)
2015	4.1 (5 211)	6.0 (10 924)	7.7 (13 306)
2016	4.8 (6 092)	4.7 (8 662)	11.5 (20 119)
2017	4.5 (5 846)	5.5 (10 230)	11.6 (20 331)
2018	4.1 (5 290)	7.0 (13 161)	13.2 (23 430)
2019	4.6 (6 002)	7.4 (14 020)	12.1 (21 599)
2020	4.1 (5 449)	7.7 (14 737)	12.9 (23 543)
2021	4.1 (5 228)	8.4 (16 262)	9.9 (18 528)
2022	4.9 (6 575)	7.5 (14 669)	8.3 (15 465)
2023	4.7 (6 410)	6.7 (13 283)	8.6 (16 092)
2024	4.7 (6 489)	5.4 (10 879)	8.0 (14 873)

Per 1 000 residents, homes completed in brackets. Administrative counts, no sampling error. Data: [cities.json](#).

Each office uses its own definition. Because Vienna's figure leaves out homes made by converting or extending buildings until 2024, Vienna is if anything understated here. Prague's figures are the Czech Statistical Office's

own per-thousand rates; its first release for 2024 gave 6 511 completed homes, revised to 6 489 in the current series¹³. Warsaw's rates use Statistics Poland's end-of-year population averaged over consecutive years, revised upward from 2020 after the 2021 census, and they match the office's published rate where one exists (11.50 and 11.56 for 2016 and 2017).

The tests use four further sources. The 2021 census round gives dwellings by the period in which their building was built for Europe's metropolitan regions, and the 2011 round, regional accounts and the metropolitan typology give the covariates^{14, 15, 16, 17, 18, 19, 20}. The Czech Statistical Office publishes permitted and completed dwellings by region every month and a quarterly price index of flats^{21, 22}. For Prague itself, the cadastre's building points give each building's flats and completion date²³, which are placed on the 1 km population grids of 2011 and 2021^{24, 25}. The Polish district panel comes from the Local Data Bank. All raw files are listed with their SHA-256 hashes.

How this was done

The study follows a registered analysis plan: design committed at 503ef06 on 29 September 2026, before the results commit 4e4301c; commit times are self-reported, and the branch was pushed after the analysis¹. The design's dated entries of 29 September, including the one headed "before any outcome was joined", were committed together with the results, so their timing rests on the author's record alone. The author has worked with these sources continuously since May 2024, in work that led to this study, so the design was written with knowledge of the sources, not blind to them; she had already published the three-city comparison and the 2024 builder split (on 28 September 2026). The plan sets the analysis in advance; the author already knew the data.

This text calls the Czech Statistical Office's "zahájené byty", literally dwellings started, permitted dwellings, because they are counted from building permits; the pipeline runs from the permit to the house number that marks completion. Three hypotheses, H2, H4 and H5, form the confirmatory family and share a 5 % family-wise error rate, at most a 5 % chance of any false positive among the three, through Holm's procedure²⁶, which tests the smallest p-value against 0.05/3, the next against 0.05/2 and the last against 0.05, and stops at the first that misses. Two others were demoted to estimates, each with a sentence fixed in advance for what would be said: the European comparison (H1), because Prague's census numerator was effectively known from the three-city count, and the price response (H3), because with one Prague and 27 quarters no valid test exists at those thresholds. Each question also has a relevance check, which tests whether the model sees the expected relationship where it should; if the check fails, the result is labelled uninformative whatever its p-value.

Power was simulated with sealed scripts that read only series from before the confirmatory windows²⁷. The simulations expected the pipeline test to be decisive at a six-month difference unless monthly noise was as large as the signal, the metro test to be decisive at a 22 % density difference, and the builders' test to be weak. The inference units are the monthly series of two regional aggregates for H2, 242 Polish districts for H4 and 524 one-kilometre squares in Prague's 22 administrative districts for H5.

Registered tests and estimates

Hypothesis	Test	Result
H2 · a permitted Prague home takes longer to get a house number than one in the rest of Czechia	Prague's mean lag minus the rest's, $\mu_P - \mu_R > 0$	supported +11.1 months, $p < 0.001$
H4 · in Poland, municipal and social builders respond less to local price growth than developers	non-market builders' price response minus developers', $\beta_{NM} - \beta_M < 0$	uninformative -0.56, $p = 0.17$; relevance check failed
H5 · in Prague, new flats are denser near the metro	change in log density per doubling of distance, $\gamma < 0$	indeterminate -0.23, $p = 0.0252$ against 0.025
H1 · Prague builds less than European peers with its demand estimate, rule fixed in advance	rank of Prague's gap from the model's prediction (conformal rank)	uninformative rank 119 of 208, $p = 0.57$
H3 · Prague's permitting responds less to price growth estimate, rule fixed in advance	90 % interval robust to serial correlation (EWC)	uninformative -11 (-86 to +63)

p-values are one-sided. H2 uses a stationary block bootstrap²⁸, which resamples runs of consecutive months, with 9 999 draws; H4 a wild score bootstrap²⁹ over 242 Polish districts, which re-weights whole districts at random to judge the estimate when observations within a district are correlated; H5 the larger of a spatial p-value (Conley³⁰, 1.5 km) and a wild score bootstrap over Prague's 22 administrative districts; H3 equal-weighted cosine (EWC) intervals³¹. Holm thresholds are 0.0167, 0.025 and 0.05. H2 passes the first and H5 misses the second, so testing stops there. H3 is the cumulative response of permitted dwellings, in log points (a log point is about 1 %), to a 1 % price rise of flats in each of the twelve preceding quarters, 2018–2024, with a price index from cadastre purchase contracts that mixes new and existing flats²²; Prague's response relative to the rest of Czechia is -11 (90 % EWC interval -86 to +63), and it is labelled uninformative because the rest's own response, 16 (-22 to +54), cannot be told from zero. Prague ranks 5th from the lowest of 13 regions; four lags over 35 quarters give +3.8 (-2.4 to +10.1), and the family-house index, a placebo, +8.5 (-25 to +42). Data: [housing_extended.json](#).

Prague among 208 European metropolitan regions

The 2021 census round recorded when each dwelling's building was built. Dwellings in buildings from 2011 or later, per resident and per year, therefore measure the last decade's new supply on nearly the same definition across 22 countries, although Czechia and Spain also count reconstructions. The unit is the metropolitan region, because housing markets do not stop at city limits, and Prague's region is Prague plus Středočeský kraj. Of Eurostat's 297 metropolitan regions, 208 have the census answer and the covariates.

Per resident, Prague's metropolitan region built 5.28 dwellings a year by the census count, which places it 163rd of 208 from the bottom, in the upper quarter on a count that includes Czech reconstructions; the median region built 3.55. Counting only the Czech Statistical Office's completions of new houses gives 3.88, mid-table at 116th of 208 on raw rates against a median of 3.50. Vienna's region built 6.2 a year and Warsaw's 10.0. The registered within-Czechia replication was run on the raw Czech Statistical Office rate instead of the model, because 14 regions cannot carry it, a deviation listed in [the change log](#): by it Prague built the second most of the 14 Czech regions per resident, 3.7 a year against a median of 2.1 elsewhere. By the census count Prague's region is in the upper quarter, and on new houses alone it is mid-table, just above the median; on both it builds less than Vienna's and Warsaw's.

The registered estimate compares each region with a prediction from what was known in 2011: population growth to 2011, size, output per head, dwellings per head, and whether the region holds a capital. The prediction for each country's regions comes from a model fitted on all other countries, and the gap is scaled by how well such regions are usually predicted^{32, 33}.

Against a demand model Prague's region ranks 119th of 208, shown as description only, since the model failed its check.

● Prague ● Vienna ● Warsaw ● other metropolitan regions

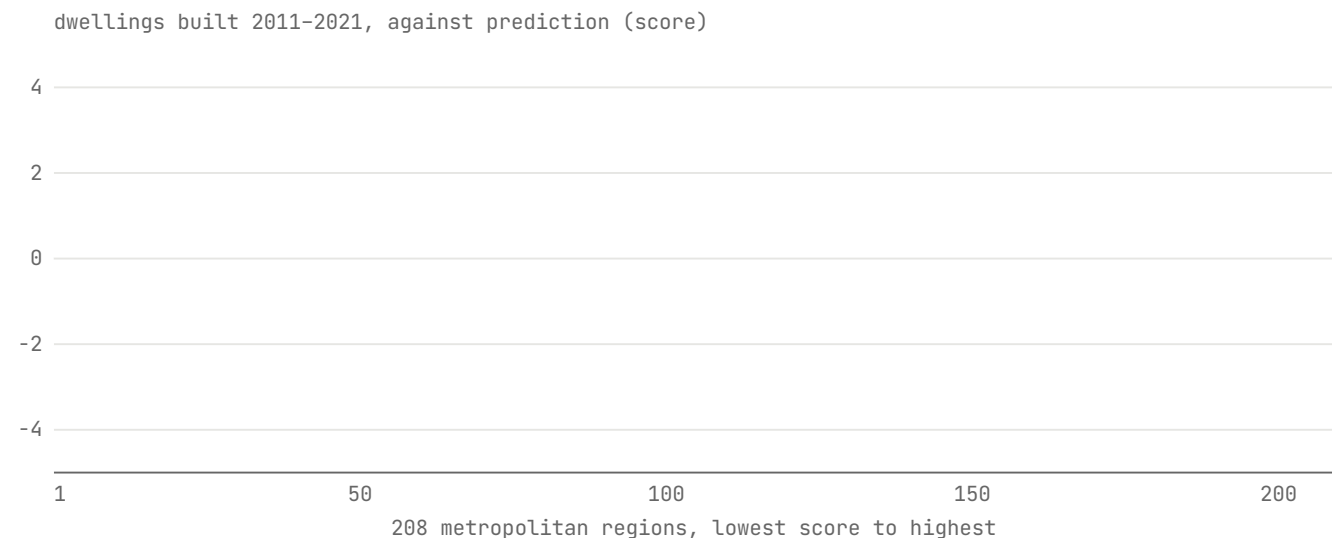


fig. 2 · Dwellings built 2011-2021 against a demand model, 208 European metropolitan regions, studentised leave-one-country-out scores (each region's gap from a model fitted without its country, scaled); zero is "as the model predicts". Prague in blue, Vienna in orange, Warsaw in olive, other regions in light grey. The model failed its relevance check, so the ranks are shown as description, not as a benchmark. Source: Eurostat census 2021 and 2011, regional accounts, accessed 28-29 September 2026.

Against that model Prague's region ranks 119th from the bottom, Vienna's 155th and Warsaw's 205th. The city on its own ranks 107th, and with the count rebuilt from the Czech Statistical Office's completions of new houses, Prague ranks 93rd as a metro region and 76th as a city. Czechia's mean score is 0.18, 8th of 22 countries from the bottom, and Prague's score lies 0.12 above the mean of the other Czech metropolitan regions, the two parts of the registered decomposition. The rule written in advance would call this "within the range of European peers", but the model failed its relevance check: regions that grew faster before 2011 did not build more afterwards, and the coefficient is negative. By the same rule the ranks are labelled uninformative.

From permit to completion in Prague and the rest of Czechia

Every month the Czech Statistical Office counts dwellings permitted and dwellings completed in each region. If Prague's flats take longer to finish, completions follow permits with a longer delay. The model estimates that delay as a distribution of lags between the two series, for flats in new apartment buildings, Prague against the other twelve regions, and the estimand is the difference between the two mean lags. Středočeský kraj is left out of the comparison, because it absorbs Prague's overflow.

In the monthly counts, completions follow permits about 11 months later in Prague (best fit; 6 to 30+ possible).

— Prague: $\hat{\mu} = 16.1$ months rest of Czechia: $\hat{\mu} = 5.1$ months (grid starts at 6) --- 95 % profile cut-off

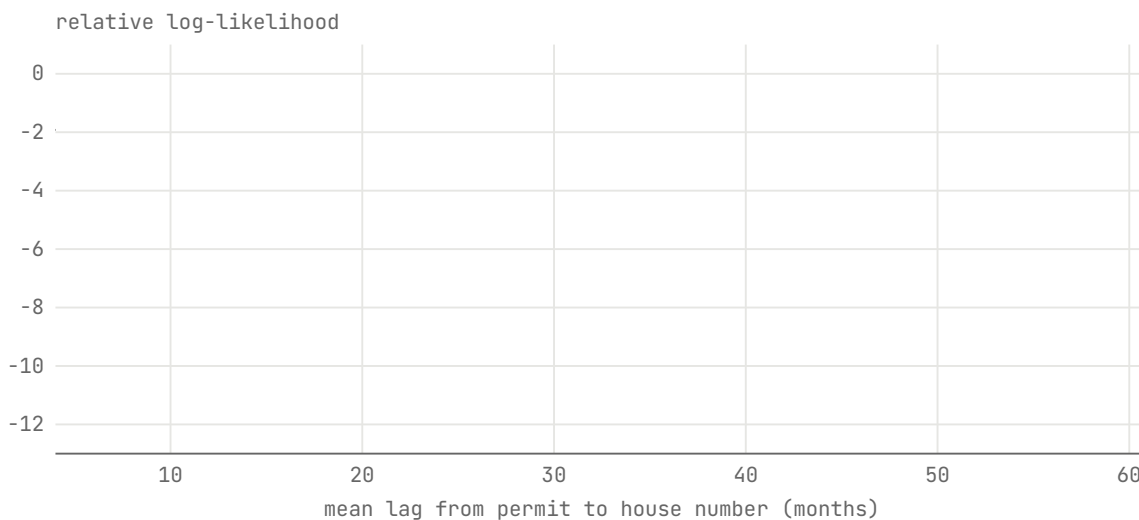


fig. 3 · How well each mean lag fits the data (relative profile log-likelihood), for Prague (blue) and the rest of Czechia without Středočeský kraj (black, dotted), on a grid of 6 to 60 months; monthly completions 2013–2024. A flat curve means the data barely distinguish between lags. Grey dashed: the 95 % cut-off. Source: ČSÚ, dwellings by region (STA09B), accessed 28 September 2026.

The best-fitting mean lag is 16.1 months in Prague and 5.1 elsewhere, a difference of 11.1 months, and H2 is supported ($p < 0.001$). The p-value uses the bootstrap standard error of 1.93 months with a normal approximation, so the interval that matches the decision is the one-sided lower bound at Holm's threshold of 0.0167, which is 6.9 months; that is, at that threshold the data rule out a difference smaller than 6.9 months. The 5th to 95th percentiles of the bootstrap draws, 8.7 to 12.6 months, are narrower than that standard error implies, because a few draws lie far from the rest.

Fig. 3 shows why both bootstrap figures understate how uncertain Prague's own lag is. The 95 % profile-likelihood region for Prague's mean lag runs from about 14 to 38 months, with one isolated grid point at 11 months also inside it, while the region for the rest of Czechia ends at about 8 months. The rest's curve peaks at the left edge of the grid, 6 months, with its estimate of 5.1 months outside the grid, so its lag is pinned at the edge and could be shorter still, and the difference inherits that. Prague's fitted lag distribution is degenerate: the model puts every flat at the single lag of 16 months, with no spread, and every bootstrap draw is refitted starting from that estimate, so the draws probably stay close to it. Read from the profile likelihood, the two regions do not overlap, so the sign of the difference holds, but its size lies somewhere between about 6 months (3 if the isolated point at 11 months is counted) and more than 30 months. The model also estimates the completion yield, the share of permitted flats that turns into completions: about a quarter in Prague (0.26) against 0.72 elsewhere, so Prague's lag rests on a weak link between the two series. In the bootstrap the yield and the lag are almost uncorrelated (-0.05), and no weight falls beyond the six-year window. Prague's monthly noise also turned out to be 3.3 times the fitted signal, beyond the design's power simulations, which stopped at a ratio of 1, where power at a six-month difference was only 25 %; a test that passes with so little power tends to overstate the size of the effect.

What the difference measures also needs qualifying. Five months is too short to build a block of flats, so outside Prague the fit probably also tracks when completions are reported relative to permits, and Prague's projects may be larger, in which case part of the lag reflects project size rather than procedure. The 11 months are therefore a

difference in the recorded pipeline, from permit to house number, which may partly be reporting timing, and not a clean difference in construction time. The result also says nothing about how long the permit itself took, which is the part of the process Czech reform has targeted⁷.

Who builds: Vienna, Warsaw and 380 Polish districts

A quarter of Vienna's new homes came from limited-profit builders; in Warsaw the public sector built 0.6 %.

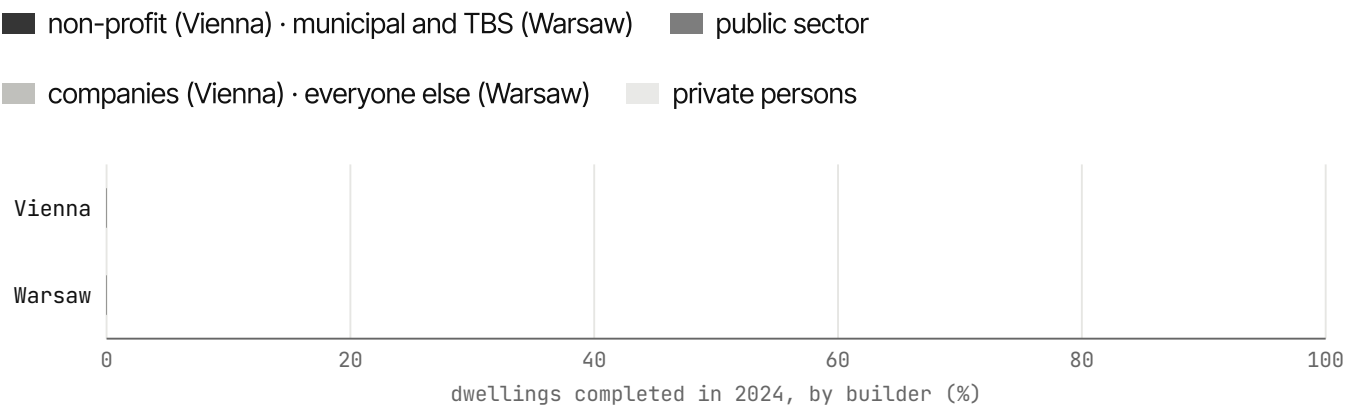


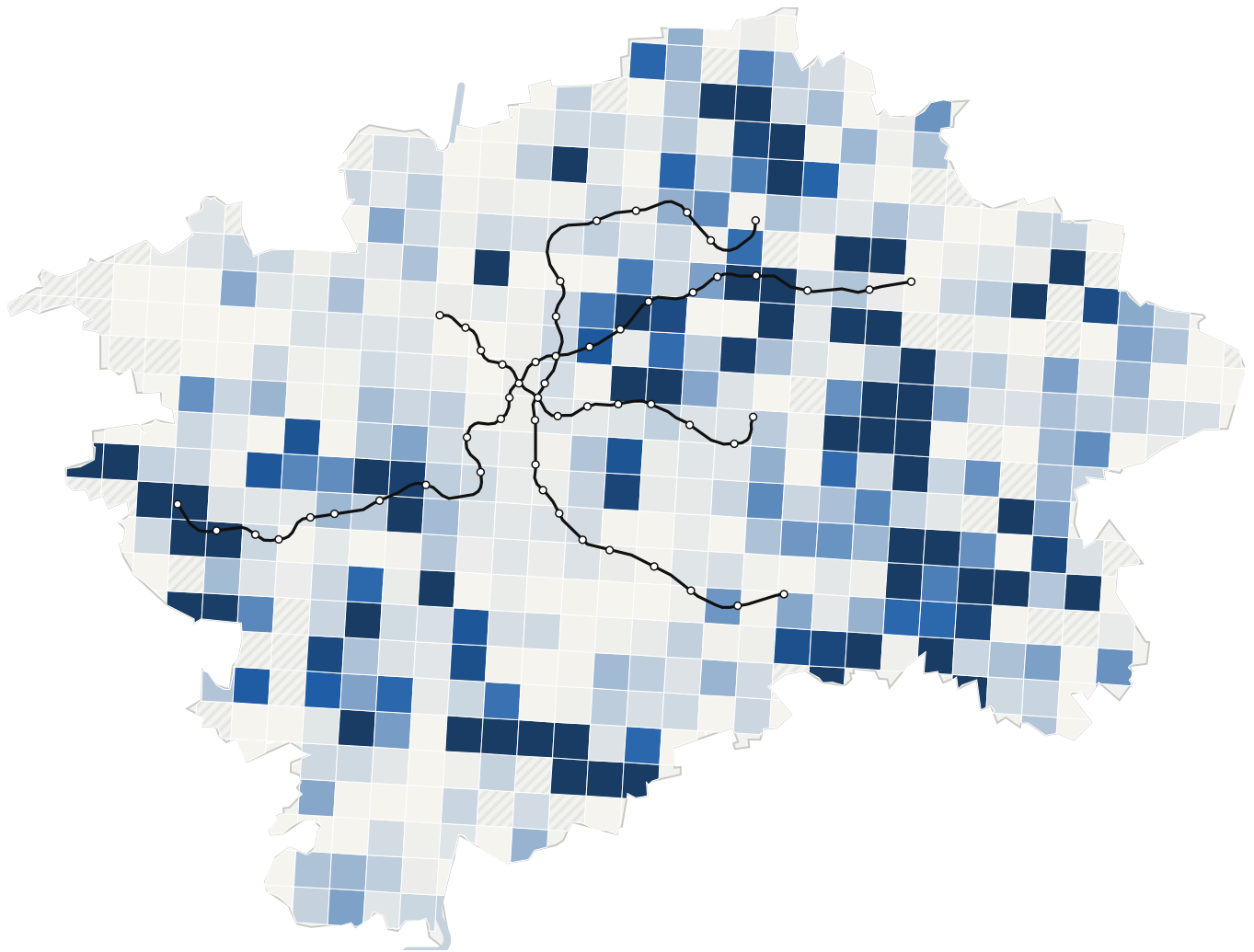
fig. 4 · Share of dwellings completed in 2024 by builder, in %; administrative counts, Vienna n = 12 478, Warsaw n = 14 873. Vienna: limited-profit housing associations (black), public sector (dark grey, 2.4 %), companies (light grey), private persons (palest grey, 6.8 %). Warsaw: municipal homes and public building societies (TBS, black) against everyone else (light grey). Sources: Statistik Austria, Statistics Poland, accessed 28 September 2026.

In 2024 a quarter of Vienna's new homes, 3 107 of 12 478, were built by limited-profit housing associations, companies whose profits Austrian law restricts, and another 2.4 % by the public sector itself. In Warsaw the city built 96 homes that year and its public building societies none, 0.6 % of the year's 14 873. The Czech Statistical Office does not publish Prague's completions by builder, so for Prague the comparison stops at volume.

No yearly Austrian series by builder was found, but Poland records, for each of its 380 districts, the dwellings that municipalities, social building societies and developers begin each year, so the registered question concerns Poland only: whether non-market builders respond less to local price growth than developers. The point estimate goes the expected way, -0.56 (90 % interval -1.52 to $+0.39$, $p = 0.17$), with the 242 districts on common support (those where both kinds of builder could be compared) acting like about 63 independent clusters, and the design had put power at the smallest difference of interest at 19 %. A separate descriptive estimate of cyclicalities, national and without year effects, gives a difference of -2.2 (standard error 4.2), equally uninformative. The relevance check fails, because developers' own starts do not visibly respond to local price growth either (0.09, $p = 0.32$), which leaves no response for the non-market builders to be smaller than, and the result is labelled uninformative. In US studies, subsidised construction partly replaces what the market would have built^{34,35}, and the Polish test says nothing about what limited-profit associations would do in Prague.

New flats and distance to the metro in Prague

Urban economics predicts that land near a metro station is worth more and that an unconstrained market would build more densely there; the prediction motivates the test and is not itself a finding here. Near-metro squares are also largely built-out estates with little land left to build on. The test counts flats completed between 2012 and census day in March 2021 in each of Prague's 524 one-kilometre squares, holding equal the distance from the centre and how many people already lived in the square in 2011.



0 200+ · flats per 1 000 residents · darkest: 200 or more

Flats completed 1 Jan 2012 to 25 Mar 2021 in Prague's 1 km squares; RÚIAN, GEOSTAT 2011.

base: Prague's city districts from the election precincts; river © OpenStreetMap contributors (ODbL)

no data

map · Flats completed in Prague's 1 km squares from 1 January 2012 to census day, 25 March 2021, per 1 000 residents of the square in 2011; hatched: nobody lived there in 2011. Above the map the number of new flats or the distance from the square's centre to the nearest station can be picked instead, and hovering over or tapping a square gives its numbers. Black: the metro and its stations as they ran in 2012, without the 2015 extension of line A. Source: ČÚZK RÚIAN building points (accessed 29 September 2026), GEOSTAT 2011 population grid, Census 2021 1 km grid.

Fewer new flats further from the metro, about 20 % per doubling, but the registered test is indeterminate.

● main estimate ● variant, $p < 0.05$ ● variant, $p \geq 0.05$ --- smallest difference of interest

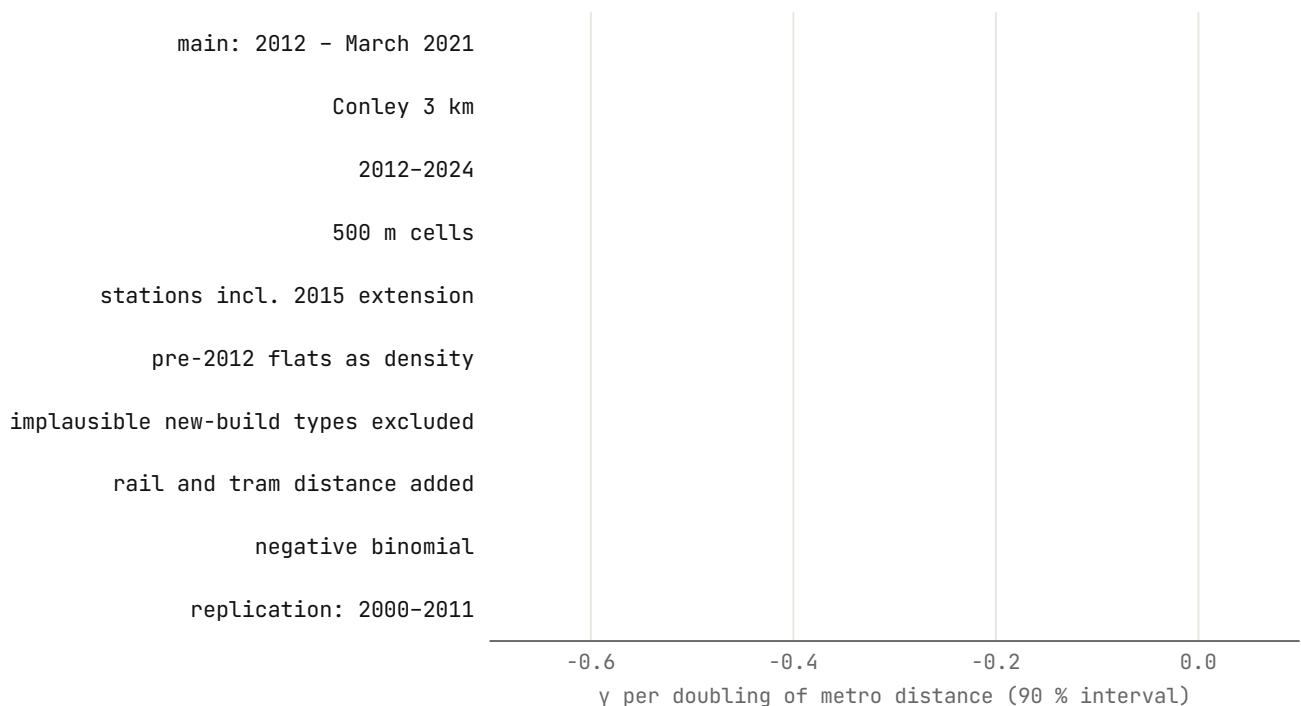


fig. 5 · Change in log new-flat density per doubling of the distance to the nearest metro station, 524 squares, with 90 % spatial (Conley) intervals; negative binomial with district-clustered intervals. Blue: the main estimate; black: variants whose decision p is below 0.05; grey: the rest. Grey dashed: the smallest effect the design cared about. Not drawn: clustering by the 57 city districts ($p = 0.015$). Source: ČÚŽK RÚIAN, accessed 29 September 2026.

The estimate is $\gamma = -0.227$ (90 % Conley interval -0.349 to -0.105), which means that each doubling of the distance to a station goes with about 20 % fewer new flats per square kilometre. This is an association: squares near stations also differ in the land left to build on, in what the city plan allows, in the age of what already stands there and in how the lines run out from the centre, and none of these was measured. The registered decision takes the larger of two p -values, 0.0011 from the spatial (Conley) standard error and 0.0252 from the wild bootstrap over Prague's 22 administrative districts, which act like fewer than 14 independent clusters. Holm's second threshold is 0.025, so H_5 does not pass, and because the reverse test does not come close either, the registered label is indeterminate. The margin, 0.0002, is within the bootstrap's own simulation error of about 0.0016, and the test was not re-run. The land and planning layers that could separate these could not be obtained in time.

Robustness

Result	Variant	Estimate
H2	Jihomoravský kraj, home of Brno, as the only comparison	+7.6 months
H2	Středočeský kraj counted as Prague	+10.3 months
H2	family houses added	+5.0 months, $p = 0.055$
H2	exponential lag instead of the registered gamma lag	+40.6 months, bootstrap diverged, no p
H2	without the COVID months	+11.2 months, bootstrap diverged, no p
H5	half-kilometre squares, 2012–2024, implausible types excluded, rail and tram added, 57 city districts	negative in every variant (fig. 5 and its caption)
H5	dropping any one district	–0.27 to –0.18
H5	2000–2011, before the window	–0.06, uncertain
H1	seven variants of the demand model (numerator, sample, covariates, weights)	45th of 116 to 138th of 208
H3	Warsaw, notified starts of works, not tested	–5.7, 4th of 376 districts

Specification curve	specifications	same sign	$p < 0.05$	robust by the registered rule
H2 · permit to house number	9	100 %	67 %	yes
H4 · builders, Poland	8	88 %	13 %	no
H5 · the metro	8	100 %	88 %	yes

A specification curve reruns the same test under each registered variant. Registered rule: robust if at least 75 % of specifications share the sign and at least 50 % reach unadjusted $p < 0.05$; the two diverged pipeline variants count as failures. H5 is robust by this rule and still missed its Holm threshold of 0.025, so its label stays indeterminate. Robustness bootstraps use 999 draws. Data: [housing_extended_robust.json](#).

The sign of the pipeline difference holds in every variant that was run, but its size depends on the assumed lag shape: 11 months with the registered gamma lag, 41 with an exponential one. The metro gradient stays negative in all of them, although in 2000–2011 it was about a quarter as steep. Against the demand model, which failed its relevance check, Prague stays out of both tails in every variant.

What this means

In official monthly counts of flats in new apartment buildings, completions follow permits with a best-fitting mean lag 11.1 months longer in Prague than in the rest of Czechia: supported ($p < 0.001$), with a one-sided lower bound of 6.9 months at Holm's threshold. The sign holds, but the profile likelihood allows a size from about 6 months to more than 30 months. The gap may partly reflect reporting timing and project size; the design identifies no cause, so it does not show that procedure after the permit delays Prague's flats, or what shortening it would change. It excludes the time taken to obtain a permit, the step the 2021 building act addressed.

Prague completed 4.5 homes per 1 000 residents a year over 2015–2024; by the 2011–2021 census count, which in Czechia includes reconstructions, its metropolitan region built 5.28 against a European median of 3.55, and on new houses alone 3.88, 116th of 208. Whether that is low for its demand is uninformative: the demand model failed its relevance check. The metro gradient, –0.23 in log density per doubling of distance (90 % Conley interval –0.349 to –0.105), is indeterminate and describes an association only.

What this does not show

The European comparison is descriptive and establishes no cause, and its demand model failed its relevance check. The pipeline result starts at the permit, so it does not measure the time it takes to get one, and the

difference it measures may partly be reporting timing and project size. Because prices respond to supply as well, the price questions measure responsiveness rather than an elasticity. The builders' test is Polish and says nothing about Prague. The metro gradient is an association, and near-station squares differ in land and planning in ways that were not measured.

The cadastre keeps only standing buildings and dates the building rather than the flats added to it; the census also dates buildings and, in Czechia and Spain, counts their reconstruction; one-kilometre squares and fourteen regions are coarse units, and results can change with the choice of areal unit³⁶.

Reproduction

The scripts live in `tools/praha/`. Raw downloads are cached in `tools/data/praha4/` and `tools/data/praha4x/`, which are not committed, and are checked against `docs/research/prague-housing-files.sha256`. Each script is run with `uv` and lists its dependencies in its header; they are not yet pinned to versions. The registered estimates are rebuilt with `DATA=tools/data uv run --with pandas --with numpy --with scipy --with statsmodels --with pyarrow --with openpyxl --with shapely --with pyproj --with pyfixest --with arch python tools/praha/housing_extended.py`, and the scripts run in this order:

1. `cities.py`: the three-city series and the 2024 builder split, `cities.json`;
2. `housing_fetch.py`: raw files and their hash list;
3. `housing_data.py`: the design frames, with no outcome summarised;
4. `housing_power.py`: design-stage power, `prague-housing-power.json`;
5. `housing_extended.py`: the registered estimates, `housing_extended.json`;
6. `housing_robust.py`: robustness, `housing_extended_robust.json`;
7. `housing_addenda.py`: the checks added after the fact-check;
8. `figures_housing_extended.py`: the figures.

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